

QP Code : NP-19812

(3 Hours)

[Total Marks : 80]

- N.B. : (1) Question No. 1 is compulsory.
 (2) Solve any **three** questions from remaining questions.
 (3) **Draw** suitable **diagrams** wherever **necessary**.
 (4) Assume suitable **data**, if **necessary**.

1. (a) Design a DFA to accept strings over the alphabet $\Sigma = \{a, b\}$ containing even number of 'a's. 5
 (b) Let G be the grammar. Find the leftmost derivation, rightmost derivation and parse tree for the expression a^*b+a^*b 5

$$G: S \rightarrow S + S \mid S * S$$

$$S \rightarrow a \mid b$$
 - (c) Give formal definition of a Push Down Automata (PDA) 5
 - (d) State and explain closure properties of regular languages. 5
2. (a) Design a DFA to accept 10
 - (i) Binary strings in which every 0 is followed by 11
 - (ii) Strings over the binary alphabet that do not contain the substring 010
- (b) Design a Mealy machine over the alphabet $\{0, 1\}$ which outputs EVEN, ODD according to the number of 1's encountered as even or odd. 10
3. (a) (a) Using pumping lemma prove that the following language is not regular 10

$$L = \{ ww \mid w \in \{0, 1\}^* \}$$
 - (b) Design a NFA for accepting input strings that contain either the keyword 000 or the keyword 010 and convert it into an equivalent DFA. 10
4. (a) Construct a PDA accepting the following language $L = \{a^n b^m a^n \mid m, n \geq 1\}$ 10
 (b) Design a Turing machine to recognize the language $L = \{a^n b^n a^n \mid n \geq 1\}$ 10

5. (a) Explain algorithm for the conversion of a Context Free Grammar (CFG) to Chomsky Normal Form (CNF) and use it to convert the following CFG to CNF 10
- $S \rightarrow bA \mid aB$
 $A \rightarrow bAA \mid aS \mid a$
 $B \rightarrow aBB \mid bS \mid b$
- (b) Convert the following Context Free Grammar to GNF 10
- $S \rightarrow AB \mid BC$
 $A \rightarrow AB \mid a$
 $B \rightarrow AA \mid CB \mid b$
 $C \rightarrow a \mid b$
6. Write short notes on (any two) 20
- (a) Variants of a Turing Machine
(b) Post Correspondence Problem
(c) Chomsky Hierarchy
(d) Recursive and recursively enumerable languages.